

DANISH NADEEM

✉ dnadeem100@outlook.com  <https://www.linkedin.com/in/dnadeem100/>

Portfolio Website: <https://d-nadeem.github.io/portfolio/>

Education

Ontario Tech University
Mechatronics Engineering

B.Eng.

Experience

Lincoln Electric

2022 - 2023

R&D Engineering Intern

East York, Toronto

- Collaborated with engineering teams on mechanical systems, electrical systems, and applying testing methodologies to prototypes and final products.
- Conducted field testing and performance evaluations of prototype machinery, documenting findings used for engineering reports and review.
- Supported prototype servicing and performed diagnostics, developing hands-on troubleshooting and maintenance skills.
- Collaborated on engineering drawings by creating drawings for new product and revising existing documents for clarity and accuracy.
- Contributed to cost-saving projects through design revisions and material efficiency improvements.
- Developed detailed SOPs that enabled team members to independently conduct test without needing prior knowledge or assistance.

Projects

Smart Airport Cart |

Capstone Project – Public Navigation System

- Led a team of four to design and implement a marker-based indoor navigation system to guide elderly users through airport terminals.
- Developed a touchscreen-based interface for user feedback and location instruction, focusing on clarity and accessibility.
- Maintained the project within budget constraints with the use of decision matrices, soliciting feedback from supervisor, other teams, and experts, reducing unnecessary expenses.
- Presented to a public audiences during the capstone exhibition, highlighting strong communication and public-facing technical explanation.

Ball Balancing using a 3RRS Parallel Manipulator |

RT Embedded Systems Project

- Achieved project success by leading the team to have the manipulator balance the ball in a live demo. Strategically divided tasks based on team members' skill sets and reallocated responsibilities to overcome challenges
- Reduced project costs compared to the reference project by optimizing hardware selection and program functionality
- Improved touchscreen program response time and reduced computational load by implementing direct port manipulation, moving average and median filters
- Completed the most challenging project in the class, as commented by the professor, assigning tasks based on skill, and utilizing resources from previous projects

2D and 3D Octopus Arm End Effector | *Onshape*

Personal Project

- Redesigned an octopus-inspired robotic arm, simplifying the structure and reducing component count to cut manufacturing cost and assembly time
- Converted the TPU-based design to PLA, increasing compression strength, load capacity, and print reliability on standard FDM printers
- Improved serviceability by developing a modular, maintenance-friendly design that reduced repair time and increased system robustness

Palletization Work Cell |

Robotics and Automation Project

- Collaborated with a team to design and develop a palletizing workcell focusing on compliance with CSA Safety Regulations CSA Z434 and CSA Z432
- Documented and validated safety protocols through CAD assemblies and risk assessments
- Ensured compliance with safety regulations by reviewing and applying CSA standards for industrial robotic manipulators

Technical Skills

Programming Experience: Raspberry Pi, STM32CubeIDE, Arduino IDE, C++, Python, OpenCV

Developer Tools: Solid Edge, NX, Solid Works, Drafting, Visual Studio Code, Onshape, Excel, Word, LaTeX